Language steering in latent space to mitigate unintended code-switching

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1 Introduction

Multilingual large language models (LLMs) frequently exhibit *unintended code-switching* - generating tokens in languages other than the target despite explicit monolingual instructions. Ryan et al. Ryan et al. [2024] document how LLM alignment procedures can cause models to default to English even when prompted in other languages, while Yoo et al. Yoo et al. [2024] demonstrate that code-switching serves as an effective red-teaming vector, bypassing safety filters. In deployed systems, such behavior degrades user trust. For instance, a customer-service chatbot prompted in Spanish may switch to English mid-response, confusing users. Existing solutions require costly fine-tuning or introduce latency via decoding-time interventions.

We propose **latent space language steering**, a cheap inference-time method that exploits low-dimensional *language directions* in hidden representations. By applying PCA to parallel translations, we identify linear subspaces separating languages while preserving semantics. Steering via simple projection (one dot product and vector subtraction per token) mitigates code-switching with negligible overhead.

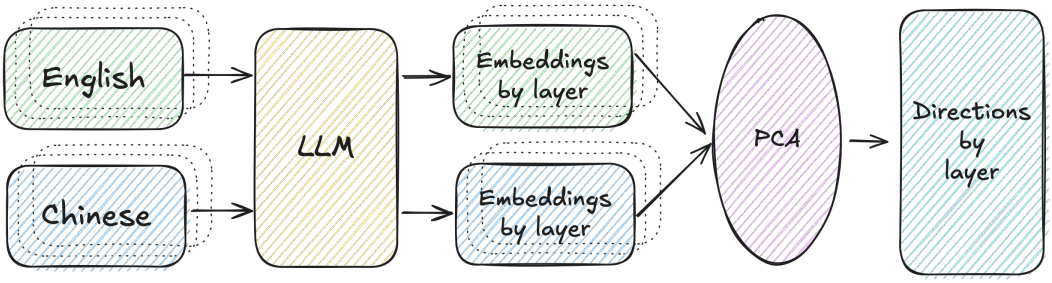
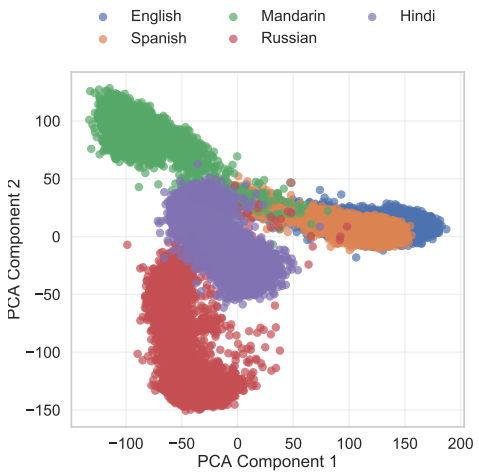


Figure 1: Language steering in hidden space: 1. Collect parallel translations of the same concepts; 2. LLM forward pass to get embeddings by layer; 3. Perform PCA by layer on the merged embeddings.

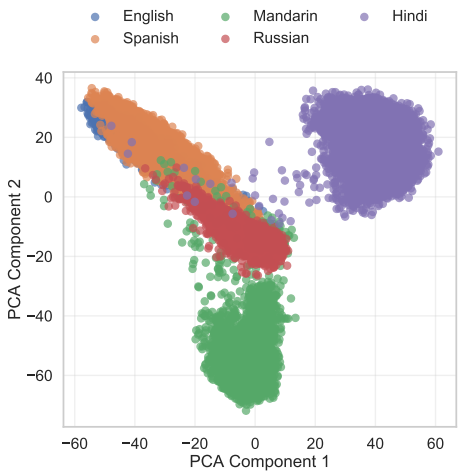
Our analysis shows language identity concentrates in final layers, with PC1 capturing most of the variance and enabling near-perfect language classification. We validate across Qwen2.5 and Llama-3.2 models on multiple language pairs (English, Spanish, Russian, Chinese, and Hindi), demonstrating reduced distributional divergence and substantially lower code-switching rates in generated text.

Code and data are released to facilitate further research [1].

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(a) Final layer language map (Qwen)



(b) Final layer language map (Llama)

2 Results

We validate our approach on Qwen2.5-1.5B Qwen et al. [2025] and Llama-3.2-1B Grattafiori et al. [2024].

2.1 Language direction identification

We use Flores Plus NLLB Team et al. [2024] dataset with 50 parallel samples for PCA fitting and 100 for validation. Figure 2a and 2b show final-layer embeddings projected onto the first two PCs. Consistent with prior findings Conneau et al. [2020], Wendler et al. [2024], models organize identical content into tight, language-specific clusters with clear boundaries.

A simple logistic regression classifier trained only on the first principal component achieves near-perfect accuracy (0.96-1.00) across all language pairs and both models.

2.2 Language steering

We fit the steering module on Flores Plus with 200 parallel samples and test on 2,467 TED Talks samples with artificial code-switching.

Table 1 shows steering reduces KL divergence by 25-55% across language pairs and models (38% average for Qwen, 43% for Llama).

Language pair	Qwen	Llama
English - Chinese	10.15 $\rightarrow$ 4.52	9.05 $\rightarrow$ 4.77
English - Spanish	7.25 $\rightarrow$ 4.90	6.52 $\rightarrow$ 3.48
English - Russian	8.70 $\rightarrow$ 5.40	7.77 $\rightarrow$ 3.49
English - Hindi	9.60 $\rightarrow$ 7.02	8.06 $\rightarrow$ 6.13

Table 1: KL divergence (no steering $\rightarrow$ steering)

2.3 Generation-based evaluation

To complement the distributional analysis above, we evaluate steering on actual text generation using Llama-3.2-1B. Across all four language pairs (English-Chinese/Spanish/Russian/Hindi), steering vectors lower the Code-Switching Index (CSI) from 0.59-0.81 (unsteered) to 0.01-0.23 while raising Target Language Consistency (TLC) to 0.77-0.99, with all differences significant at $p < 0.001$.

[1] https://github.com/fxlrnrpt/language-steering-in-latent-space

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